Name: Samiksha B N

SRN: PES1Ug24CS413

Section: 5G

Code coverage

```
(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6>coverage erase

(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6>coverage run -m pytest CodeCoverageTest_daycare_fees.py
==================== test session starts ====================
platform win32 -- Python 3.13.15, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6
collected 12 items

CodeCoverageTest_daycare_fees.py ............            [100%]

==================== 12 passed in 0.14s =====================

(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6>coverage report -m
Name                               Stmts   Miss  Cover   Missing
----------------------------------------------------------------
CodeCoverageTest_daycare_fees.py      28      0   100%
daycare_fees.py                       37      3    92%   20-22
----------------------------------------------------------------
TOTAL                                 65      3    95%

(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6>
```

Branch coverage

```
(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees>cd "Lab 6"

(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6>coverage erase

(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6>coverage run --branch -m pytest BranchCoverageTest_daycare_fees.py
==================== test session starts ====================
platform win32 -- Python 3.13.15, pytest-9.2.0, pluggy-1.7.0
rootdir: C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6
collected 46 items

BranchCoverageTest_daycare_fees.py ..................... [ 45%]
.........................                                [100%]

==================== 46 passed in 0.21s =====================

(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6>coverage report -m
Name                                 Stmts   Miss Branch BrPart  Cover   Missing
------------------------------------------------------------------------------
BranchCoverageTest_daycare_fees.py      21      0      0      0   100%
daycare_fees.py                         37      0     26      0   100%
------------------------------------------------------------------------------
TOTAL                                   58      0     26      0   100%

(venv) C:\Users\Samiksha\L6_L7_48_daycare_fees\Lab 6>
```

Test Cases

Test Case 1: Sibling discount for a non-infant child

- Function: extra_hours_fee
- Test file: CodeCoverageTest_daycare_fees.py and BranchCoverageTest_daycare_fees.py
- Input: age_group = “preschool”, hours = 40, has_sibling = True
- Expected output: 45
- Covers: the True branch of if has_sibling and age_group != "infant", and the non-infant rate of 9.

Test Case 2: Surcharge for more than 20 extra hours (infant)

- Function: extra_hours_fee
- Test file: BranchCoverageTest_daycare_fees.py
- Input: age_group = “infant”, hours = 41, has_sibling = False
- Expected output: 462
- Covers: the True branches of if extra > 20 and if age_group == "infant" inside it (lines 19-22). The paired test with 40 hours (extra = 20, fee 280) covers the boundary where the surcharge does not apply.

Test Case 3: Enrolment boundary with immunisation pending

- Function: enrolment_allowed
- Test file: BranchCoverageTest_daycare_fees.py
- Input: age_months = 12, vacancies = 0, is_immunisation_pending = True
- Expected output: True
- Covers: the True branch of if is_immunisation_pending on the exact boundary. The paired test with age 11 months returns False, covering the other side.

Test Case 4: Pickup status for aftercare on a Friday

- Function: pickup_status
- Test file: BranchCoverageTest_daycare_fees.py
- Input: minutes_left = 14, is_aftercare = True, is_friday = True
- Expected output: “overdue”
- Covers: the True branches of if is_aftercare and if is_friday, and the final fall-through to “overdue” after the if and elif conditions.